Emergency 911 Call Counts in Negative Binomial Distribution

Statistical Analysis Module 2 Project

This project studies the daily number of Montgomery County emergency 911 calls in each one-hour period. The sample variance is larger than the sample mean for the hourly counts, so a Poisson model is not suitable here. We use a negative binomial model and compare its cumulative probabilities with the data.

Assignment Requirements

The assignment asks for a clear random variable, its expectation and variance, and results using a cumulative distribution. We define daily hourly call counts, calculate their means and variances, and compare empirical and negative binomial tail probabilities $P(X >= k)$.

Data and Study Design

The source data contain 663,522 Montgomery County 911 call records from 10 December 2015 at 17:10:52 to 29 July 2020 at 15:52. Each call was assigned to its calendar date and hour. For every year and hour, the number of calls was counted for each day.

To avoid comparing partial days with full days, 10 December 2015 and 29 July 2020 were excluded. The analysis uses 21 days in 2015, 366 days in 2016, 365 days in 2017 through 2019, and 210 days in 2020. The boundary years are not full calendar years and should be interpreted cautiously.

Random Variable and Model

Let $X(y,h)$ be the number of 911 calls on one day in year y during hour h, where $h = 0, 1, ..., 23$. For a fixed year and hour, repeated daily values of $X(y,h)$ form the sample used for estimation. The sample mean μ estimates $E[X]$, and the sample variance s^2 estimates $Var(X)$.

Why the Poisson Distribution Was Not Used

A Poisson random variable has $E[X] = Var(X)$. Figure 1 compares the daily mean μ and the sample variance s^2 for all 24 hours in each year. In all 144 year-hour cells, s^2 is larger than μ. The difference is especially large in several afternoon hours. This is called overdispersion. It is the main reason for not using the Poisson distribution.

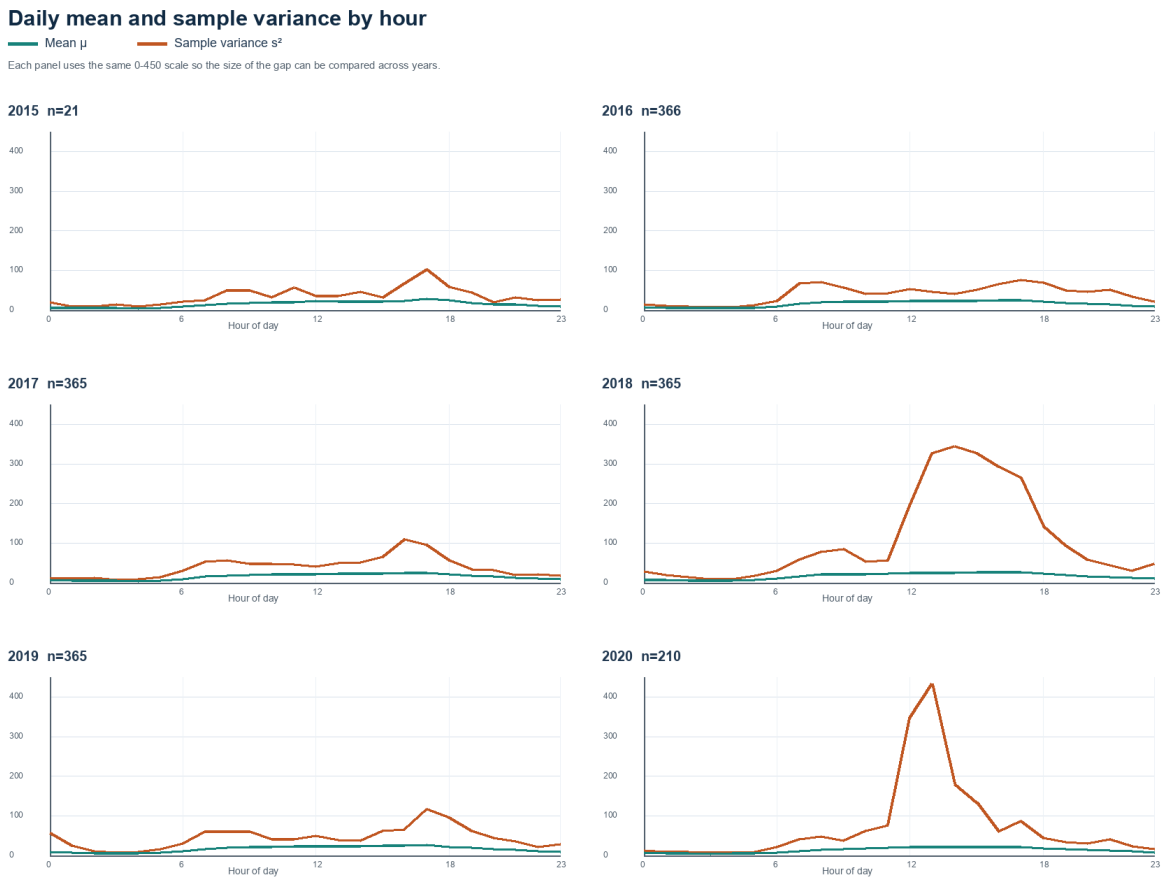


Figure 1. Daily mean μ and sample variance s² for the 24 hours in each year

Negative Binomial Distribution

The negative binomial distribution can handle Var(X) > E[X]. In the parameterization used here, X counts failures before r successes and p is the success probability.

$$P(X = x) = \frac{\Gamma(x + r)}{\Gamma(r)\Gamma(x + 1)} p^r (1 - p)^x$$

$$r = \frac{\mu^2}{s^2 - \mu}, \quad p = \frac{\mu}{s^2}, \quad P(X \geq k) = 1 - F(k - 1)$$

For an integer threshold k, the report compares the empirical tail probability, the proportion of days with at least k calls, with the fitted negative binomial tail probability. The Excel calculation is 1 - BETA.DIST(p, r, k, TRUE). This is equivalent to 1 - F(k - 1), where F is the negative binomial cumulative distribution function.

Hourly Call Pattern

Call activity is lowest overnight, rises after 06:00, and is highest in the afternoon and early evening. The pattern is similar in the four complete years.

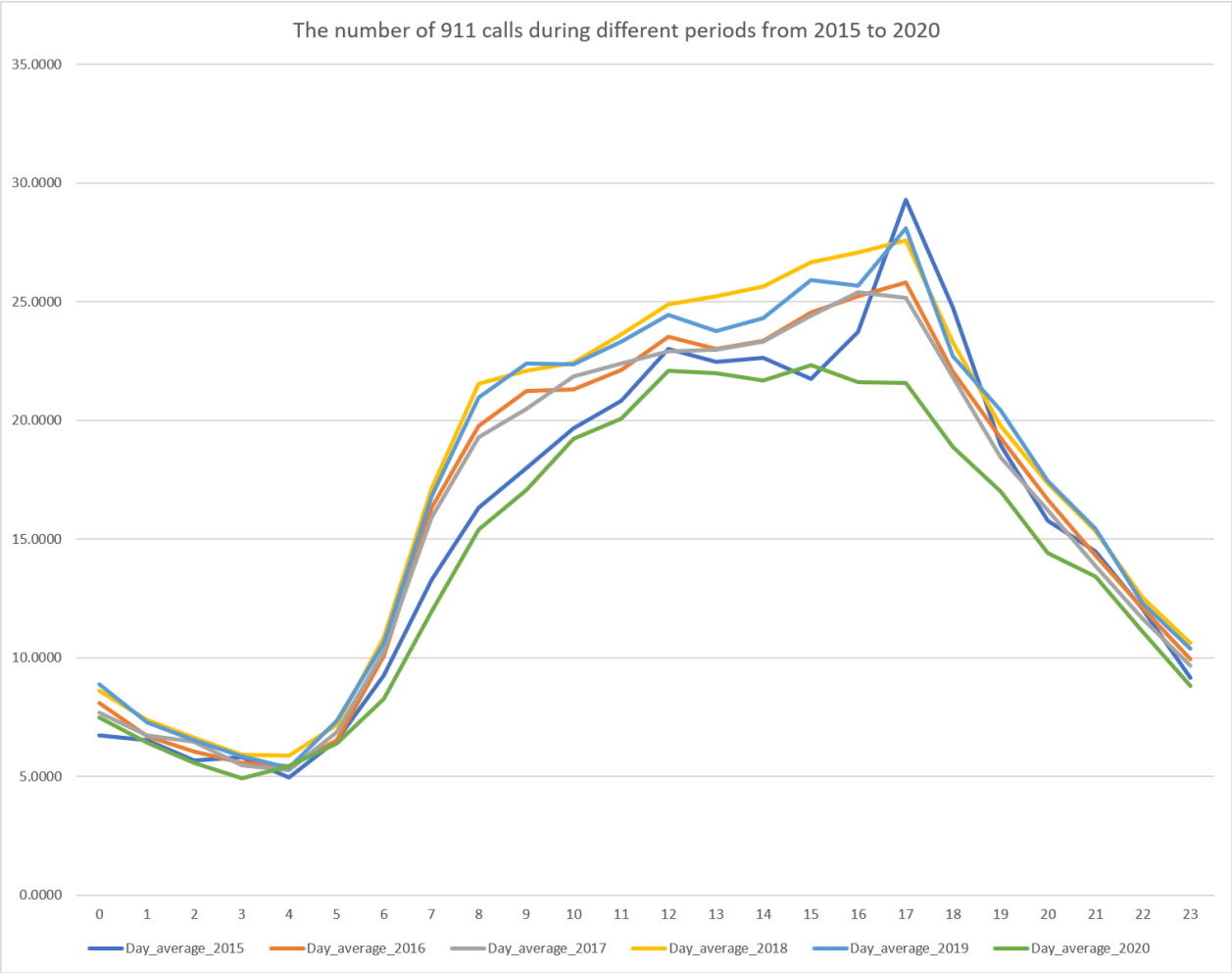


Figure 2. Mean daily 911 calls by hour and year

The 2020 curve is lower at many hours. A reasonable possible explanation is the change in travel and daily activity during COVID-19. However, the 2020 data cover only 1 January to 28 July, so this chart cannot prove that COVID-19 caused the decrease.

Frequency Groups and Selected Thresholds

The 24 hours were split into low, middle, and high frequency groups according to their usual daily mean. The checking points k are different for each group because a useful range depends on the usual number of calls. These points are selected nodes for checking the cumulative distribution, not every possible value of k.

Frequency group	Hourly mean μ	Thresholds k used for checking
Low	about 5 to 10	3, 6, 9, 12, 15
Middle	about 10 to 20	5, 10, 15, 20, 25, 30
High	about 20 to 30	10, 15, 20, 25, 30, 35, 40

Cumulative Probability Results

For each selected threshold, the empirical probability is the proportion of days with at least k calls. It is compared with the fitted negative binomial probability. The table reports the fitted parameters and the results for four complete-year 10:00-11:00 panels. It also includes 2018 13:00-14:00 as a higher-variance example.

Period	n	Mean μ	Variance s^2	r	p	MAE
2016 10:00-11:00	366	21.31	41.12	22.93	0.518	0.59%
2017 10:00-11:00	365	21.84	48.65	17.80	0.449	0.89%
2018 10:00-11:00	365	22.41	54.08	15.86	0.414	1.61%
2019 10:00-11:00	365	22.36	41.49	26.12	0.539	0.98%
2018 13:00-14:00	365	25.21	327.37	2.10	0.077	13.65%

For 10:00-11:00, the empirical and fitted tail probabilities are close. The MAE ranges from 0.6% to 1.6%. In contrast, 2018 13:00-14:00 has a much larger variance and an MAE of 13.7%. This shows that the negative binomial model is useful for many hours, but it does not fit every hour equally well.

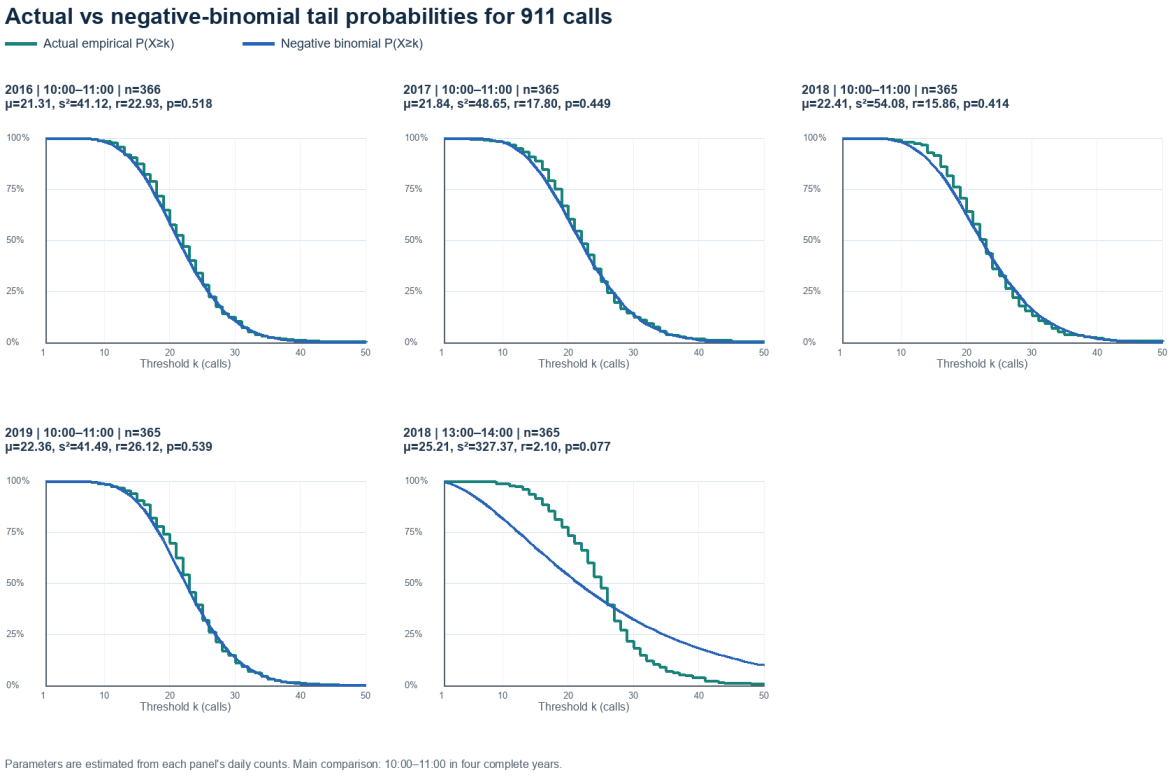


Figure 3. Empirical and fitted negative binomial tail probabilities P(X >= k), k = 1 to 50

Error Summary

For one year and one hour, each selected threshold gives an absolute difference: |Actual P(X >= k) - NB P(X >= k)|. MAE is the average of these differences. It is only an approximate summary because it uses the selected checking nodes in the table. Figure 3 is more detailed because it uses every integer k from 1 to 50 for its five examples.

The largest MAE values occur in some midday and afternoon periods, especially in 2018 and the partial 2020 sample. These periods may contain clustering or unusual days that a two-parameter moment fit does not fully describe.

Hour	2015	2016	2017	2018	2019	2020
0:00	5.58%	2.39%	0.83%	5.56%	10.96%	0.49%
1:00	4.38%	1.73%	0.73%	5.97%	7.81%	0.32%
2:00	2.41%	0.85%	0.58%	0.88%	1.43%	1.06%
3:00	3.64%	1.93%	1.16%	1.06%	0.71%	0.78%
4:00	3.08%	1.76%	1.67%	0.98%	2.70%	1.06%
5:00	2.69%	2.48%	2.47%	3.27%	2.72%	1.10%
6:00	5.21%	0.59%	3.32%	1.55%	1.97%	2.17%
7:00	4.25%	2.31%	1.14%	1.49%	1.40%	1.99%
8:00	3.05%	1.12%	1.50%	1.43%	0.55%	1.04%
9:00	4.36%	1.27%	1.28%	4.22%	1.61%	1.70%
10:00	1.25%	0.58%	1.21%	2.18%	1.72%	2.98%
11:00	4.82%	0.63%	0.61%	1.97%	0.87%	5.23%
12:00	3.68%	1.11%	1.05%	11.05%	2.31%	16.28%
13:00	3.16%	0.79%	1.00%	15.97%	0.88%	17.45%
14:00	6.43%	0.66%	1.08%	15.97%	0.74%	9.68%
15:00	2.73%	1.05%	2.35%	13.32%	0.47%	8.42%
16:00	2.29%	0.80%	4.72%	12.98%	1.65%	2.16%
17:00	3.72%	1.14%	4.07%	10.26%	3.26%	2.51%
18:00	2.08%	2.47%	1.98%	8.74%	6.22%	0.56%
19:00	3.71%	1.87%	0.98%	6.86%	4.28%	1.49%
20:00	3.29%	4.76%	1.86%	5.34%	2.01%	1.81%
21:00	3.49%	5.97%	0.29%	3.55%	1.76%	3.38%
22:00	2.18%	3.04%	0.55%	3.16%	1.09%	2.12%
23:00	3.84%	3.19%	2.45%	7.70%	4.44%	2.58%

Figure 4. Mean absolute error for selected negative binomial tail probabilities

Conclusion and Limitations

Daily 911 call counts within an hour are discrete random variables. Their means and variances show overdispersion, so the negative binomial distribution is more suitable than the Poisson distribution. The cumulative probability comparison is close for several complete-year 10:00-11:00 samples.

The fit is weaker for some high-variance periods. The negative binomial distribution is therefore a useful approximation, not a perfect model for every hour. The 2015 and 2020 samples are partial years. A later study could test weekday, season, and call-type effects.

References. Montgomery County 911 Calls. Kaggle dataset. https://www.kaggle.com/datasets/mchirico/montcoalert. Coursework calculations and figures were prepared from Original_data.csv and the processed Excel files for this project.